22/01/25 m/s

USN | | | | | | | | | | [23ESEE12]

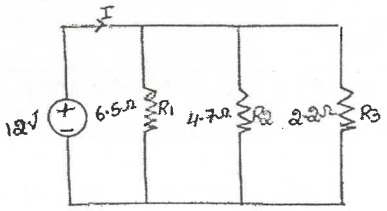
FIRST SEMESTER B.Tech EXAMINATION, JANUARY - 2025

BASICS OF ELECTRICAL & ELECTRONICS ENGINEERING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

			Blooms Level	CO's	Marks
		PART - A			**5x2=10**
1.	**a.**	Define Kirchoff's Current Law and Kirchoff's Voltage Law.	L1	CO1	**02**
	b.	What are the advantages of AC system over DC system?	L2	CO2	**02**
	c.	List the different types of cables.	L1	CO3	**02**
	d.	Find out the Value of resistor for the following colors (First Three Bands Colors): Grey, Red and Orange.	L1	CO4	**02**
	e.	Define the terms: Avalanche Breakdown.	L1	CO5	**02**
		PART - B			**6x5=30**
2.	**a.**	For the network shown in figure, calculate the effective resistance and supply current. 	L3	CO1	**05**
		OR			
	b.	An alternating current i is given by i = 141.4 sin 314t Find **i.** The maximum value **ii.** Frequency **iii.** Time Period **iv.** The instantaneous value when t=3ms	L3	CO1	**05**
3.	**a.**	Derive the expression for Voltage, Current and Power for a Single-phase AC circuit containing Pure resistance.	L3	C02	**05**
		OR			
	b.	Define RMS Value and derive the expression for RMS value of a Sinusoidal Voltage.	L2	C02	**05**
4.	**a.**	With help of neat diagram, Explain Pipe Earthing and Plate Earthing.	L2	CO3	**05**
		OR			

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	b.	Explain: **i.** Switch Fuse Unit (SFU) **ii.** Miniature Circuit Breaker (MCB)	L2	CO3	**05**
5.	**a.**	Describe briefly polarized and non-polarized electrolytic capacitors.	L2	C04	**05**
		OR			
	b.	Explain the different types of Inductors.	L2	C04	**05**
6.	**a.**	Explain the block Diagram of regulated Power supply with associated diagram.	L2	CO5	**05**
		OR			
	b.	Explain the operation of an electronic Instrumentation system with a neat block diagram.	L2	CO5	**05**
7.	**a.**	Explain the V-I Characteristics of PN Junction Diode.	L2	C06	**05**
		OR			
	b.	Illustrate the circuit of the Half Wave Rectifier and explain its operations with waveforms.	L2	C06	**05**
		PART - C		**6x10=60**	
8.	**a.**	Define series resistor circuit? Explain its characteristics and voltage distribution in series circuit.	L2	CO1	**10**
		OR			
	b.	Show that the equivalent resistance of a group of resistors connected in Parallel is the reciprocal of the sum of the reciprocal of each of resistance.	L2	CO1	**10**
9.	**a.**	A pure inductive coil allows a current of 10A to flow from a 230V, 50 Hz supply. Find **i.** inductance of the coil **ii.** Power absorbed and **iii.** equations for voltage and current.	L3	CO2	**10**
		OR			
	b.	Illustrate with neat diagram and explain the RLC Series Circuit and phasor diagram with different cases.	L3	CO2	**10**
10.	**a.**	What is battery? Explain the different types of batteries.	L2	CO3	**10**
		OR			
	b.	Define Switch. Discuss in brief on the different types of switches used in Electrical circuits.	L2	CO3	**10**
11.	**a.**	Illustrate the Elementary calculations for energy consumption with necessary formula. **i.** A household has the following meter readings: 12300 (1 December 2016); 13200 (28 February 2017). The previous energy bill was: 3960 kWh (from 10 November 2015 to 5 November 2016).	L3	CO4	**10**

		ii. A vacuum cleaner is used for two hours per day and 52 days in a year. The capacity of vacuum cleaner is 2000 watts. Calculate the energy consumption of the vacuum cleaner.			
		OR			
	b.	Show the average power consumed by a inductance is zero. Explain with suitable equations.	L2	CO4	**10**
12.	**a.**	Explain the working of simple Zener Diode voltage regulator.	L2	CO5	**10**
		OR			
	b.	Derive an expression for **i.** PIV **ii.** V_{dc} &I_{dc}. **iii.** V_{rms} & I_{rms} **iv.** Ripple Factor	L2	C05	**10**
13.	**a.**	Illustrate the principle of operation of PN Junction diode along with its applications.	L2	C06	**10**
		OR			
	b.	Illustrate the circuit of Bridge rectifier and explain its operations with waveforms.	L2	C06	**10**